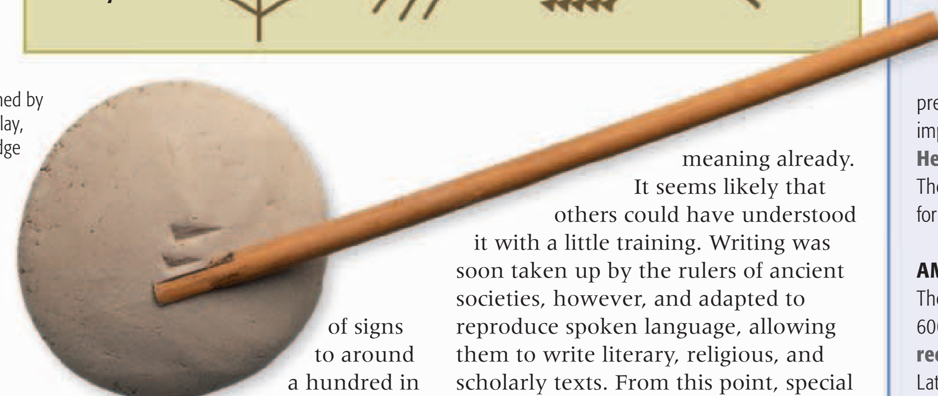
Egyptian hieroglyphs
Formal writing in Egypt retained the use of pictorial symbols—hieroglyphs—for more than 3,000 years. This example from a 4th-century-BCE sarcophagus differs little in style from the earliest surviving inscriptions made in *c.* 3200 BCE. When reading hieroglyphs, the reader starts at the top on the side the signs face (in this case, right).

FROM PICTORIAL SYMBOLS TO CUNEIFORM

The earliest cuneiform, or proto-cuneiform, was pictorial and drawn into clay. At some point, all proto-cuneiform signs were rotated by 90 degrees. No one knows why. True cuneiform appeared when scribes began to form signs from impressed wedges.

Date	3200 BCE	3000 BCE	2400 BCE	1000 BCE
GIN "to walk"				
UD "day"				
MUSEN "bird"				
SE "barley"				



Cuneiform stylus
Cuneiform signs were formed by pressing a stylus into wet clay, each time producing a wedge shape. Cuneiform means "wedge-shaped" in Latin.

Numerals are expressed with these circular impressions. They mirror the shaped clay tokens once used to signify numbers (see BEFORE). They appear next to the sign for the commodity (barley). Before the invention of numerals, scribes had to draw a sign once for each item.

pictures of the things it records. Over time, these pictures were simplified and made abstract to make writing quicker and easier. In Mesopotamia, this process resulted in wedge-based cuneiform writing (see box, above). Many early scripts were logographic, meaning that each symbol represented an entire word or idea. A logographic system may use thousands of signs. Modern Chinese writing remains logographic, using around 12,000 symbols that allow written communication between the many different dialects of Chinese. Cuneiform and Egyptian hieroglyphic scripts, meanwhile, mixed logograms with symbols representing sounds. Such sound signs were combined to form words, which reduced the total number of signs to around a hundred in scripts such as Akkadian cuneiform. Egyptian and Maya hieroglyphs remained pictorial for decorative use in religious writing and inscriptions on monuments. For everyday use, however, the Egyptians developed a more efficient, abstract system called hieratic. It was written with fragile reed pens, which restricted the shapes the scribe could form. When written on papyrus, hieroglyphs were painted with brushes, allowing the scribe a freer hand.

Chinese writing also diverged, with different styles of calligraphy being developed for different uses. In most Chinese scripts, the meaning of signs was simplified as well.

The earliest writing records only objects (usually goods) and numbers (quantities of goods and measurements of time). Grammar was absent, so this writing cannot be read as language, but it aided the memories of people who knew its meaning already. It seems likely that others could have understood it with a little training. Writing was soon taken up by the rulers of ancient societies, however, and adapted to reproduce spoken language, allowing them to write literary, religious, and scholarly texts. From this point, special training was needed.

Spread of the written word

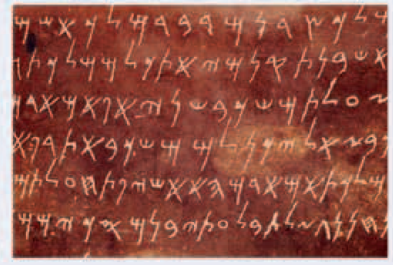
Cultures in the 3rd and 2nd millennia BCE were not really literate societies. Once writing became abstract, rather than pictorial, only a small number of merchants, administrators, and elites would have had enough schooling to read and write. It is thought that only one per cent of Egyptians were literate.

Egyptian scribe
Education of scribes began in childhood, lasting at least 10 years, and included mathematics and accountancy. The scribal profession usually ran in families.

Ancient rulers used writing to manage the information on which their states ran, not to disseminate it. Royal political inscriptions might be combined with imagery, and it seems that the masses would have read only the images, while the writing was aimed at fellow elites and at posterity. Assyrian kings, for instance, buried inscriptions in the foundations of temples, recording their exploits so that future kings rebuilding those temples would read them.

AFTER

Writing systems became simpler and more sophisticated, but the spread of written communication was slow until printing was invented during the European Renaissance.



PHOENICIAN ALPHABETIC SCRIPT

ALPHABETIC SYSTEMS

At first, written symbols represented a variety of words, syllables, ideas, or sounds. The idea that every symbol should denote a sound was an innovation in the Middle East and led to the alphabet. The **first alphabetic writing**, with each sign representing a **consonant** but with **no vowels**, appeared in the 2nd millennium BCE, using adapted Egyptian hieroglyphs. The people of **Ugarit** in Syria developed a cuneiform alphabet, but the need for clay prevented it from spreading. Alphabets became important in 1000–700 BCE, being used for **Hebrew, Aramaic, and Phoenician** writing. The Phoenicians **82–83 »** used separate signs for vowels, **influencing Greek and Latin** writing.

AMERICAN SYSTEMS

The earliest surviving American writing is on 600 BCE **Zapotec** monuments in Mexico and **records the names of sacrificed captives**. Later inscriptions on **Maya monuments** record conflicts between city-states **140–41 »**. The **cultures of the Andes developed quipu** **212 »**—a system that recorded numerical information with patterns of knots on webs of color-coded string.

ZAPOTEC CALENDAR

PRINTING

The spread of written material was hampered by the need to **copy by hand**. In Europe from 1454, with the **Gutenburg printing press** **253 »** featuring **movable type**, books were produced quickly and cheaply on a large scale.

LETTERS OF MOVABLE TYPE READY FOR PRINTING